

OLLITAL

TECHNICAL WHITE PAPER

Particle Engineering for Improved Performance

Connecting feed properties, atomization, drying kinetics and powder collection to measurable product performance.

[/solutions/particle-size-control/](#)

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Decision Brief

Particle size is the result of connected decisions. Feed preparation influences droplet formation, drying controls morphology, and the collection train determines which particles reach the final container. A useful development plan links every adjustment to a measurable powder response.

A repeatable operating window is more valuable than one ideal sample produced under a narrow condition.



WHAT THE REVIEW SHOULD DECIDE

01 Start with product function

Define the downstream behavior the powder must deliver before choosing a numerical size target.

02 Control the feed state

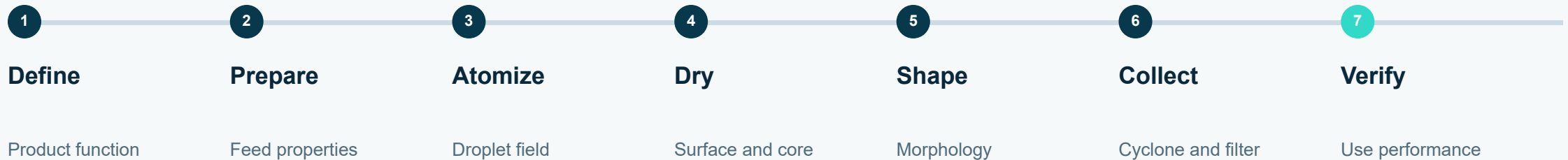
Solids, viscosity, dispersion and temperature influence atomization before the nozzle or wheel starts.

03 Report the recovered PSD

Particle size data should be interpreted together with yield, moisture and collection losses.

Particle Formation Path

The particle distribution develops across the complete process. A change at one stage can move size, density, morphology and recovery at the same time.



Atomization Technology Comparison

Atomizer selection defines the main adjustment mechanism. The useful range must be confirmed with the real feed and the intended powder specification.



Two-fluid nozzle

Best fit: flexible laboratory work

Gas-to-liquid ratio and feed rate provide a broad adjustment range at lower throughput. Review gas demand and nozzle wear.



Pressure nozzle

Best fit: scalable liquid pressure

Orifice and swirl geometry create a simple spray path. Viscosity, plugging and wear can narrow the operating window.



Rotary atomizer

Best fit: broad turndown

Wheel speed controls mechanical energy independently from feed pressure. Wheel design and chamber geometry remain linked.

Droplet Formation Directions



OLLITAL reviews the atomizer as part of the chamber, air system and powder recovery train. Hardware direction follows the material behavior and scale-up evidence required.

01 Feed preparation

Control dispersion, temperature, solids and viscosity before atomization.

02 Atomization energy

Select a stable adjustment range that tolerates normal feed variation.

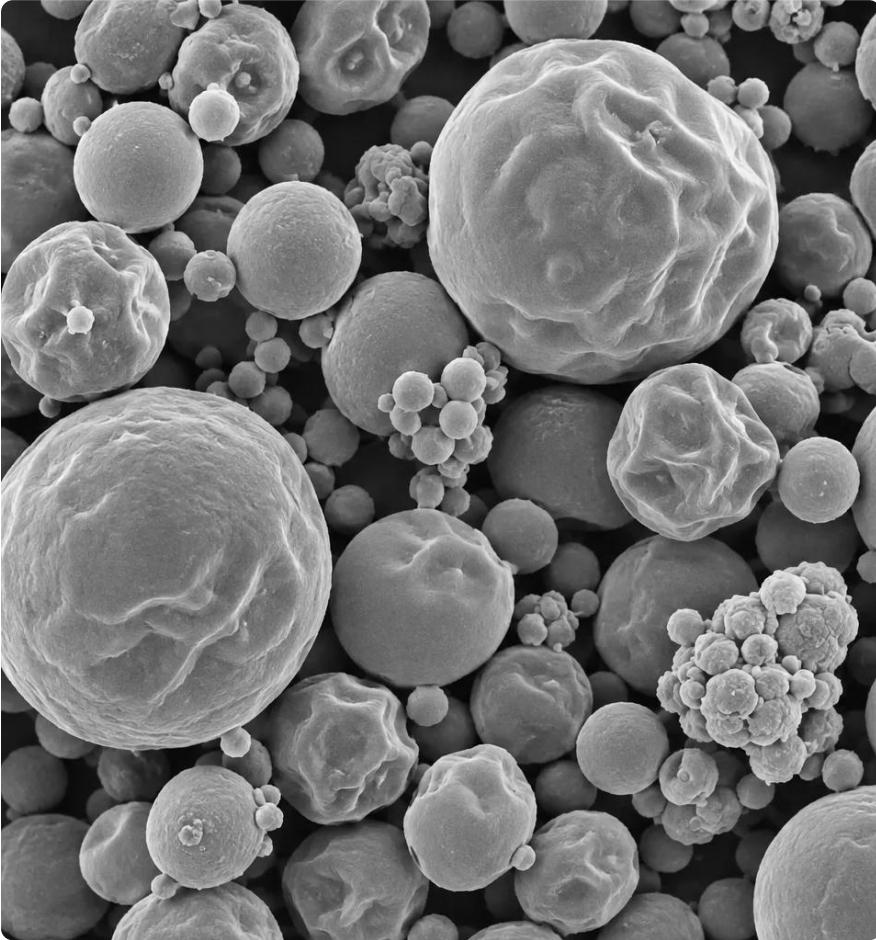
03 Drying kinetics

Balance surface drying, internal diffusion and wall interaction.

04 Collection cut

Quantify cyclone and filter fractions so fines do not disappear from the analysis.

Particle Response Variables



Directional effects help design experiments, but the final response depends on the material and the complete equipment configuration.

Atomization energy

Often moves characteristic droplet size and the recovered fines fraction.

Feed solids

Changes viscosity, droplet mass, density and the drying load applied to each particle.

Drying intensity

Influences shell formation, shrinkage, porosity, moisture and wall deposition.

Collection behavior

Cyclone and filter cut points can shift the measured recovered distribution.

Project Input Checklist

A clear target reduces trial-and-error. Provide the downstream function and the measurement method beside the numerical particle specification.



Product function

- Final dosage or use
- Flow behavior
- Dissolution target
- Density requirement

Feed properties

- Solids and viscosity
- Dispersion state
- Surface tension
- Temperature stability

Particle target

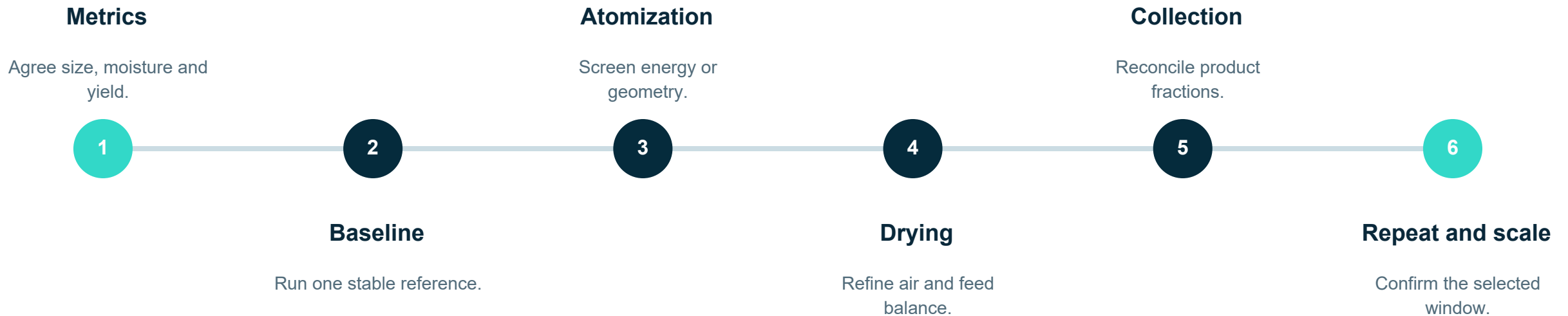
- D10, D50 and D90
- Span or fines limit
- Morphology preference
- Moisture target

Measurement

- PSD instrument
- Dispersion method
- Microscopy
- Yield basis

Structured Development Program

The program changes one engineering family at a time and records powder quality together with operating stability.



Record the complete equipment configuration with every sample. A powder result without nozzle, air, feed and collection context cannot support scale-up.

Development and Measurement Framework

Particle specifications should support product performance and use a documented measurement method. The same framework should continue from laboratory work into scale-up.

01 Quality attributes

Connect particle size, morphology, density, moisture and yield to the intended product function.

02 Analytical method

Document sample preparation, dispersion settings, instrument configuration and reporting basis.

03 Representative scale-up

Preserve the mechanisms that control atomization, drying and collection rather than volume alone.

04 Change control

Treat formulation, atomizer, collection hardware and analytical method changes as meaningful.



Turn the powder target into a test plan

Share the feed properties, target distribution, downstream function, available sample quantity and project stage. OLLITAL can define a practical laboratory or pilot route.

MATERIAL EVALUATION

<https://ollitaldryer.com/material-evaluation/#evaluation-form>

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